Breast Cancer Tumor Board: A Radiologist's Guide to Postmastectomy Radiation Therapy

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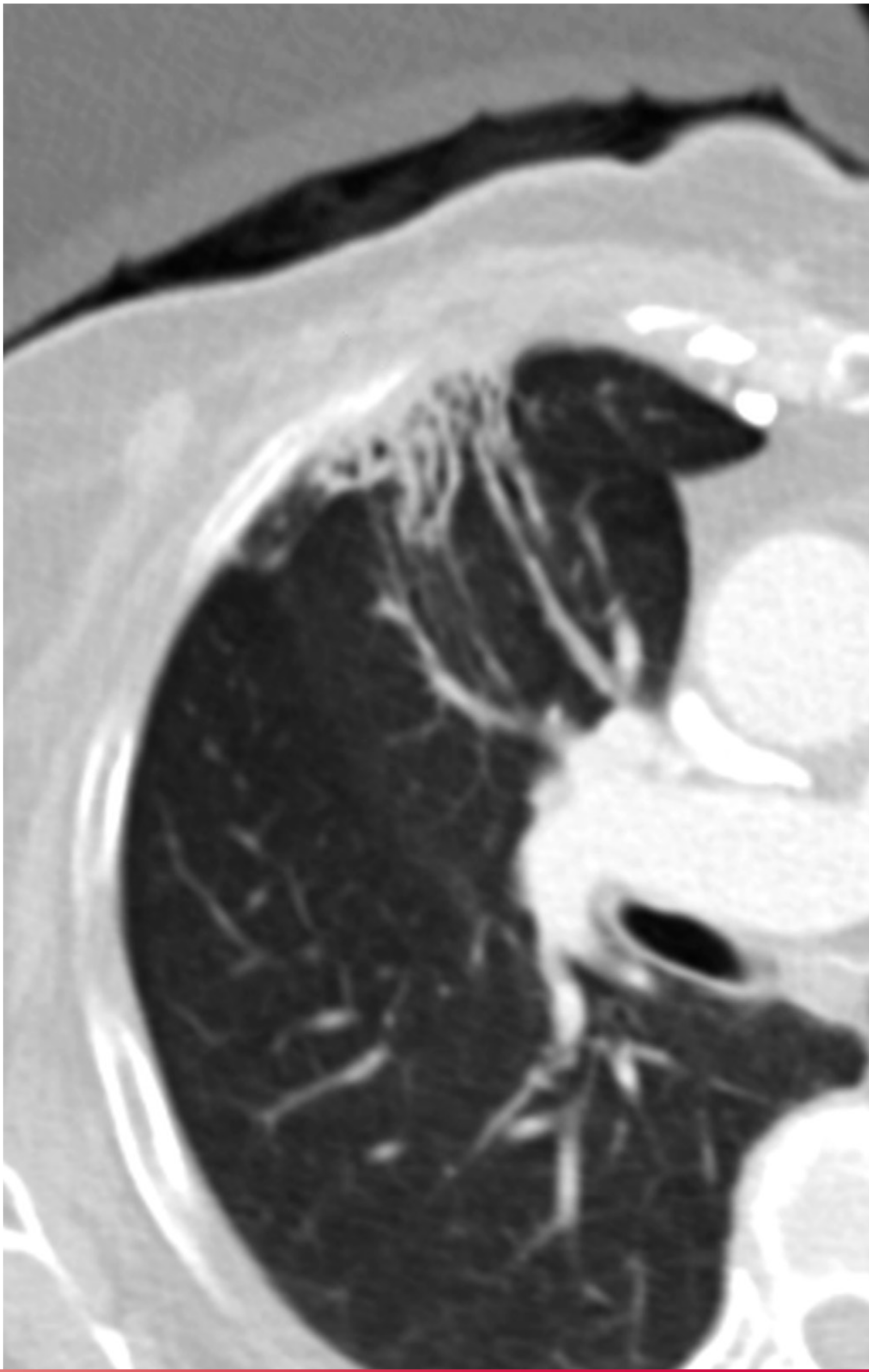
Radiation therapy represents a pillar in the current management of breast cancer. Historically, postmastectomy radiation therapy (PMRT) has been administered only in patients with locally advanced disease and a poor prognosis. These included patients with large primary tumors at diagnosis and/or more than three metastatic axillary lymph nodes. However, during the past few decades, several factors have prompted a shift in perspective, and recommendations for PMRT have become more fluid. Guidelines for PMRT in the United States are outlined by the National Comprehensive Cancer Network and the American Society for Radiation Oncology. Because evidence to support performing PMRT is frequently discordant, the decision to offer radiation therapy often requires team discussion. These discussions are usually held in multidisciplinary tumor board meetings in which radiologists play a pivotal role by providing critical information such as the location and extent of disease. Breast reconstruction after mastectomy is optional and is safe in cases in which the patient's clinical status allows it. The preferred method in the setting of PMRT is autologous reconstruction. If this is not possible, then a two-step implant-based reconstruction is recommended. Radiation therapy does involve a risk of toxicity. Complications can be seen in acute and chronic settings and range from fluid collections and fractures to radiation-induced sarcomas. Radiologists have a key role in detecting these and other clinically relevant findings and should be prepared to recognize, interpret, and address them.



Supplemental Material

Quiz questions for this article are available in the supplemental material.

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Abbreviations: ALND = axillary LN dissection, ASTRO = American Society for Radiation Oncology, BCT = breast conservation therapy, IMN = internal mammary node, LN = lymph node, MTB = multidisciplinary breast tumor board, NAST = neoadjuvant systemic therapy, NCCN = National Comprehensive Cancer Network, PMRT = postmastectomy radiation therapy, 3D = three dimensional

TEACHING POINTS

- At MTBs, radiologists provide critical information for decision making. Familiarity with the guidelines for and controversies surrounding PMRT ensures the ability of radiologists to contribute relevant accurate information to the care team.
- Current indications for mastectomy are broad and include factors such as genetic propensity, young age and premenopausal status, large tumor size relative to breast size (as lumpectomy cosmesis is unlikely to be satisfactory), many cases of multicentric or multifocal disease, chest wall involvement, contraindications to postlumpectomy breast radiation therapy, inflammatory breast cancer (stage T4d), persistently positive margins following BCT, and recurrence following radiation for BCT. In some cases, mastectomy may be indicated in patients with Paget disease.
- Essentially all PMRT treatments consist of ionizing radiation delivered with photons or charged particles (protons, electrons) by way of an external beam. Photon (x-ray) treatment can be delivered by using either three-dimensional (3D) conformal or intensity-modulated radiation therapy techniques.
- In certain clinical scenarios, NCCN and ASTRO recommendations diverge. As a result, many cases are not clear cut and the recommendations ultimately depend on the physician's judgment. Therefore, there may be a benefit for clinical decision making from multidisciplinary discussions in the setting of breast cancer MTBs.
- Complications from radiation therapy may be acute or chronic and may not be seen until decades after the treatment. Complications range from mild dermatitis to rare but potentially fatal radiation-induced second malignancies. They may be classified as early (occurring weeks to months after radiation therapy), intermediate (occurring months to years after radiation therapy), or late (occurring more than 10 years after radiation therapy).

Introduction

Radiation therapy is a mainstay of current breast cancer management. The addition of adjuvant whole-breast radiation to lumpectomy has allowed breast conservation therapy (BCT) to surpass mastectomy as the preferred treatment for early-stage breast cancer while maintaining excellent oncologic outcomes (1). Nevertheless, up to 40% of patients with breast cancer will still undergo mastectomy (2).

Traditionally, postmastectomy radiation therapy (PMRT) has been administered only in cases of locally advanced disease with a poor prognosis (3). These included patients who had large primary tumors at diagnosis and/or more than three metastatic axillary lymph nodes (LNs). Concern for treatment-related toxic effects such as secondary malignancy, lymphedema, and lung and cardiac disease in particular has limited the more liberal use of PMRT.

During the past few decades, several factors have prompted a shift in perspective. First, reports in the literature showed that a multitude of individual patient and tumor characteristics affect the postmastectomy prognosis and that even patients with less advanced disease can potentially benefit from PMRT (4,5). At the same time, the oncologic community recognized that some patients may not benefit from PMRT owing to a low risk of local-regional recurrence, comorbidities, and/or limited life expectancy (3). For example, Luo et al (6) analyzed the findings in patients with early breast cancer who were treated with and without radiation therapy after mastectomy, while characterizing their risks. They concluded that patients with high risk clearly benefit from PMRT, while those with low risk should undergo PMRT benefit-risk evaluation before they receive radiation therapy. Moreover, improvements in systemic therapy, particularly that targeting human epidermal growth factor receptor 2, have called into question the benefits of PMRT in subsets of patients. These factors forged a new approach to PMRT that involves balancing risks (3) and individualizing care. As a result, recommendations for PMRT have become more fluid. Radiologists continue to play a critical role in the decision to offer PMRT, as this decision is sometimes dictated entirely on the basis of imaging findings such as the presence of unresected gross lymphadenopathy.

Furthermore, it became clear that the management of patients requiring PMRT is extraordinarily complex. As PMRT is most effective when it is part of trimodality therapy (surgery, radiation, and chemotherapy) (7), it requires fastidious coordination with chemotherapy and, if desired, breast reconstruction procedures. In addition, risk acceptance and tolerance of toxic effects vary among patients and physicians. For these reasons, current guidelines recommend that the choice to administer PMRT be made by means of clinical consensus, preferably at a multidisciplinary breast tumor board (MTB) (7,8). At MTBs, radiologists provide critical information for decision making. Familiarity with the guidelines for and controversies surrounding PMRT ensures the ability of radiologists to contribute relevant accurate information to the care team.

Multidisciplinary Breast Tumor Board

An MTB is a regularly scheduled meeting of medical specialists who combine their expertise to discuss the optimal therapy for patients with cancer. These boards usually consist of medical oncologists, radiation oncologists, pathologists, surgeons, radiologists, nurse navigators, and others. Multidisciplinary discussions in clinics or cancer tumor boards are now recommended by most experts and in most guidelines (9). A 2013 study from the American Society of Clinical Oncology showed that 85% of the responding clinical practices participated in MTBs (9,10).

Although their influence has been studied to a limited extent (11), MTBs are purported to add substantial value to cancer care. For example, clinicians rely heavily on MTBs for guidance in diagnosing cases and managing patients with cancer. In a 2013 study by Charara et al (9), 86.7% of cases at the American University in Beirut were presented at an MTB to determine patient management. As a result of these discussions, management alterations were made in 67% of cases. MTBs are believed to be particularly helpful in complex cases (10). Based on the evaluation of 15 recorded MTBs, Schellenberger et al (12) concluded that "atypical" cases evoke more discussion and therefore more benefit from the MTB than do routine cases. Other potential benefits of tumor boards include standardizing practice patterns, ensuring appropriate adherence to expert guidelines, improving communication between providers, providing learning opportunities, and improving patient accrual in clinical trials.

Despite their presumed benefits, MTBs have not been shown to unequivocally improve patient care. A 2013 study to examine the outcomes of patients with cancer (10) whose cases were reviewed at tumor boards within a large Veterans Affair health system showed no benefit in standardization of care, cost of services, or survival. This result may have been due to a number of factors that may plague MTBs, including the variable expertise of the MTB participants and leaders, intermittent absence of key disciplines, and lack of feedback loops that inform clinicians of patient outcomes. Schellenberger et al (12) also found that the effectiveness of communication in MTBs is dependent on the availability of standardized guidelines for reference.

Nevertheless, MTBs have become common practice in the management of patients with cancer, and participating radiologists offer essential information at every step of patient care—namely, diagnosis, treatment, planning, and surveillance. In addition, MTBs enable radiologists to more fully participate in patient care and educate their colleagues as well as themselves.

Indications for Mastectomy

Surgery is indicated for essentially all localized breast cancers. Different surgical approaches have been developed throughout the years, with a preference for more conservative surgeries when the extent of disease allows. Mastectomy generally involves an elliptical incision and excision of all mammary tissue while preserving blood supply to the overlying skin (13).

There are several types of mastectomies, which can be classified as partial (ie, lumpectomy) (14), simple, modified radical, and radical. Other variations in technique include skin-sparing mastectomy and nipple-areola–sparing mastectomy, procedures that often accompany breast reconstruction (15). Nipple-sparing mastectomy generally is oncologically appropriate for treatment of in situ carcinoma and stage I–II invasive cancers. Previously, this technique was recommended only in cases in which tumor was located 2 cm or more from the nipple (15,16). However, newer studies (17) have shown that nipple-sparing mastectomy may be effective in cases in which tumor is located within 1 cm of the nipple—and even when tumor is within 5 mm when there is confirmation that the nipple base is free of disease.

Current indications for mastectomy are broad and include factors such as genetic propensity, young age and premenopausal status, large tumor size relative to breast size (as lumpectomy cosmesis is unlikely to be satisfactory), many cases of multicentric or multifocal disease, chest wall involvement, contraindications to postlumpectomy breast radiation therapy, inflammatory breast cancer (stage T4d), persistently positive margins following BCT, and recurrence following radiation for BCT. In some cases, mastectomy may be indicated in patients with Paget disease (13).

Genetic Predisposition

Prophylactic bilateral mastectomy also may be recommended for patients who are diagnosed with breast cancer and have an increased lifetime risk of developing malignancy in the contralateral breast. This includes patients with germline genetic mutations in genes such as *BRCA1* or *BRCA2*, *TP53*, and *PTEN;* patients who have a strong family history (18); and patients who are younger than 35 years or premenopausal at diagnosis (8). In patients who have unilateral disease with average risk, routine contralateral prophylactic mastectomy does not provide a significant survival benefit and is discouraged (15).

Pregnancy

Because pregnancy is an absolute contraindication to any type of radiation therapy, pregnant patients in their first trimester who are diagnosed with breast cancer and elect to continue the pregnancy should undergo mastectomy with axillary staging. For some patients in their second and third trimesters, the National Comprehensive Cancer Network (NCCN) recommends preoperative chemotherapy followed by mastectomy. The timing of chemotherapy during pregnancy (only indicated after the first trimester) should be coordinated with the obstetrics team to determine optimal timing. For patients in the late third trimester, treatment options include mastectomy or BCT with axillary staging, followed by postpartum radiation therapy (8).

Inability to Achieve Negative Lumpectomy Margins

Mastectomy is indicated in patients who have undergone BCT for ductal carcinoma in situ or invasive cancer with surgical margins that were persistently positive for tumor after re-excision (13). It is also indicated in certain cases of recurrence after lumpectomy in which second lumpectomy and/or adjuvant radiation therapy would not be feasible (13).

Large and Multifocal or Multicentric Disease

Breast cancer is referred to as “large” if it measures more than 5 cm (with or without associated calcifications), in part because 5 cm served as an eligibility criterion in certain landmark studies of PMRT, including the Danish Breast Cooperative group 82b (4) and 82c trials. In select patients with no indications for mastectomy other than large tumor size, neoadjuvant chemotherapy can be used to attempt to downstage the tumor and permit BCT (15).

Multifocal disease refers to more than one tumor in the same quadrant, and multicentric disease refers to more than one tumor in multiple quadrants (19). Traditionally, multifocal disease and multicentric disease have been treated with mastectomy. However, investigators in a relatively recent study (20) assert that in this setting, there is no survival benefit to mastectomy, as compared with BCT, provided that all of the tumor can be excised with satisfactory cosmesis. Radiologists’ input in determining tumor size and multifocality exemplifies their impact on patient care.

Locally Advanced Breast Cancers and Inflammatory Breast Cancer

According to the eighth edition of the *AJCC Cancer Staging Manual*, locally advanced breast cancers are large tumors

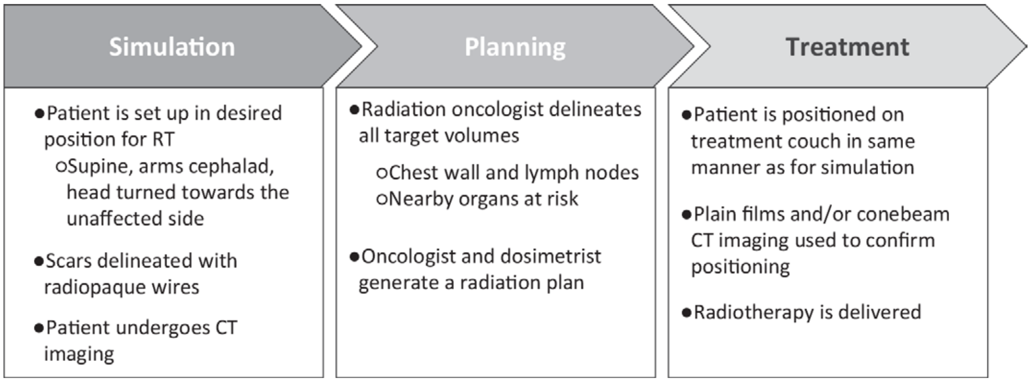


Figure 1. Diagram outlines radiation therapy *(RT)* planning steps.

Figure 2. Example of patient positioning for CT simulation. **(A)** The patient is placed supine in the CT scanner with her arms above the head. **(B)** Mastectomy scar with wire marking.

(T3 and T4) with or without involvement of the chest wall or skin and with extensive lymphadenopathy or involvement of second-line nodes (supraclavicular, infraclavicular, or internal mammary chains). Inflammatory breast cancer denotes invasion of the dermal lymphatics. These tumors are typically aggressive and are treated with mastectomy. Patients usually receive neoadjuvant chemotherapy to maximize the likelihood of undergoing mastectomy, since neoadjuvant systemic therapy (NAST) can render inoperable tumors operable (8). If the response to chemotherapy is inadequate, patients rarely may also receive neoadjuvant radiation therapy.

Rationale for PMRT

Radiation therapy is an integral component of breast cancer treatment. With few exceptions, such as in elderly patients with early-stage tumors in whom only tamoxifen therapy after surgery is a reasonable option (21,22), patients who undergo lumpectomy should receive adjuvant radiation treatment. The main benefits that support adjuvant radiation therapy after BCT are not only a reduction in local-regional recurrence but also a decreased cancer-related mortality risk at 15 years (23). However, in the setting of mastectomy, radiation therapy is not uniformly indicated. As radiation therapy, unlike systemic therapy, is a localized treatment modality, the patients who benefit most from PMRT are those who are at increased risk for local-regional recurrence. These are patients who have positive surgical margins after mastectomy, large tumors or regional LN metastases, or pathologic features that are suggestive of aggressive tumor biologic features. By preventing cancer cells from recurring and potentially metastasizing, radiation therapy added to surgery and chemotherapy improves overall survival and disease-free survival (3,8).

Fundamentals of PMRT

Essentially all PMRT treatments consist of ionizing radiation delivered with photons or charged particles (protons, electrons) by way of an external beam. Photon (x-ray) treatment can be delivered by using either three-dimensional (3D) conformal or intensity-modulated radiation therapy techniques. Proton therapy remains investigational but is increasing in frequency, as it can potentially lower the radiation dose to organs at risk, particularly in challenging cases.

The first step in radiation treatment is to perform radiation planning CT, referred to as CT simulation, with the patient in the treatment position (Fig 1). For radiation therapy to an intact breast, patients are frequently placed in the prone position. Conversely, for radiation therapy following mastectomy, patients are positioned supine, usually with the arms extended above the head to limit the radiation dose to the ipsilateral upper extremity (Fig 2). The radiation oncologist then delineates on CT images the targets and organs at risk for use in radiation planning. Reverification of the patient setup for PMRT is routinely accomplished by using orthogonal conventional radiography or cone-beam CT and is performed at least weekly.

3D Conformal Radiation Therapy

The most common radiation technique for PMRT remains photon treatment performed by using forward-planned 3D

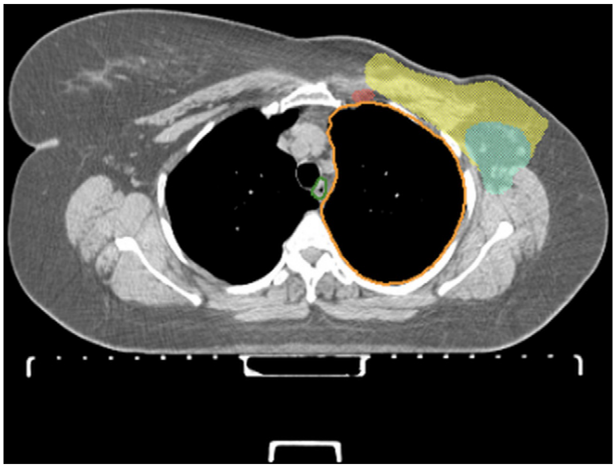


Figure 3. Volume delineation for a patient undergoing PMRT. Yellow demarcates the chest wall contour; red, the internal mammary nodes (IMNs); and teal, the axillary level I nodes. The lung is outlined in orange, and the esophagus is outlined in green.

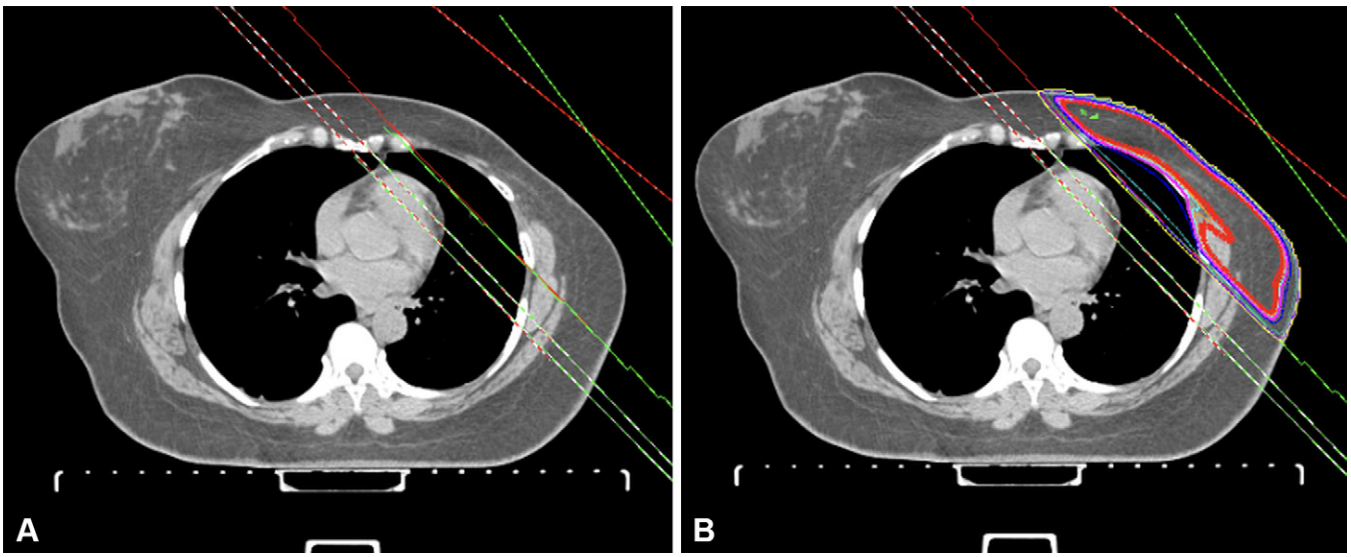


Figure 4. Typical PMRT chest wall tangent beams, with red lines denoting medial tangent beams and green lines denoting lateral tangent beams. In **B**, the thick red isodose line denotes the area receiving the prescription dose (5000 cGy) for this patient.

conformal radiation therapy (24). With forward planning, the angles and dimensions of photon beams are set to deliver the intended radiation dose to the target structures—namely, the chest wall with or without regional lymphatics—while sparing nearby organs at risk (Fig 3). Typically, two opposed tangent fields are used to treat the chest wall, level I axillary nodes, and internal mammary nodes (IMNs). The fields are placed at an angle to spare the contralateral breast and minimize the radiation dose to the heart and lungs (Fig 4).

There are several approaches to treating the level III axillary nodes, IMNs, and supraclavicular region. These approaches often vary according to treatment provider and/or institution. For example, the supraclavicular and level III axillary LNs may be treated with a single anterior oblique radiation field, angled 10°–15° from the midline to avoid the spinal cord and esophagus. This is sometimes combined with an opposed field (ie, posterior axillary boost) to better treat the deep level III axilla. If coverage of the ipsilateral IMNs is desired but cannot be achieved with use of chest wall tangents that spare the heart or lung, a separate photon or electron field may also be used (24).

Intensity-modulated Radiation Therapy

Intensity-modulated radiation therapy (IMRT) is an advanced photon radiation therapy technique that may be considered when heart or lung dose-volume constraints cannot be met with use of 3D conformal radiation therapy planning. With IMRT, target volumes and organs at risk are delineated and the desired radiation doses are set by the radiation oncologist. The computerized treatment planning system then generates an optimized radiation plan that conforms to the target structure. The system conforms to the target by way of multileaf collimators that modulate the intensity of the beam at different points (24). Consequently, IMRT typically involves a larger number of fixed beam angles compared with 3D conformal radiation therapy. Volumetric modulated arc therapy is a variant of IMRT in which the radiation dose is delivered continuously as the gantry rotates about the patient.

Proton Therapy

In contrast to photons, protons are heavy ions that deposit the maximal radiation dose at a specific distance in tissue (the Bragg peak), distal to which they deposit little additional dose. This potentially allows the dose to nearby organs to be minimized while the radiation coverage of target volumes is maintained (24). Proton therapy remains investigational for PMRT, with numerous ongoing clinical trials. Compared with photon treatment, proton therapy is potentially attractive in this setting because it allows the chest wall and IMNs to be treated with relative ease with use of anterior beams while sparing the underlying heart and lung to a greater degree. The benefits of proton therapy are offset by increased treatment time and greater susceptibility to errors due to minor day-to-day variability in positioning. In addition, proton therapy centers are limited or unavailable in many geographic regions.

Radiation Dose

PMRT is usually delivered to the chest wall and regional LNs by using conventional radiation that is fractionated to a dose of 4500–5040 cGy during 25–28 treatments (24). Mastectomy scars containing residual tumor and/or clinically involved LNs that cannot be safely resected (eg, high axillary nodes, supraclavicular nodes, and IMNs) may require additional radiation doses (boosts), to a total dose of 6000–6600 cGy.

Techniques to Minimize Cardiac Radiation Dose

It is well established that breast or chest wall radiation therapy has the potential to cause deleterious long-term effects on the heart, lungs, and other organs. The radiation dose to cardiac tissue is of particular clinical relevance, especially in cases of left-sided breast tumors (25). As screening mammography has facilitated more timely detection of breast cancer, most patients with early-stage disease can be cured (26). Therefore, mitigating late cardiac toxic effects is essential to survivorship.

CT-based planning with accurate delineation of the heart is indispensable for defining the volumetric radiation dose. While the cardiac dose can be minimized with prone positioning when treating the intact left breast, this is not possible with PMRT. When using traditional 3D conformal radiation therapy photon fields in PMRT, the heart can be protected by using a technique known as deep inspiration breath hold, in which radiation is delivered in 15–30-second intervals, during which the patient holds a deep breath. This expands the lungs and flattens the diaphragm, displacing the heart posteriorly from the chest wall. The deep inspiration breath hold can be voluntary or device assisted (27). Some have recommended the use of continuous positive airway pressure–assisted therapy, but this remains experimental (28). Use of intensity-modulated radiation therapy and proton therapy also may reduce the dose to the heart in PMRT.

Indications for PMRT

In the United States, the most widely referenced guidelines for the use of PMRT are issued by the NCCN (8) and the American Society for Radiation Oncology (ASTRO) (29). Both sets of guidelines are comprehensive, but those from the NCCN tend to be more detailed. The guideline directives range from “recommend” to “strongly consider if” to “consider if.”

In certain clinical scenarios, NCCN and ASTRO recommendations diverge. As a result, many cases are not clear cut and the recommendations ultimately depend on the physician’s judgment. Therefore, there may be a benefit for clinical decision making from multidisciplinary discussions in the setting of breast cancer MTBs (10,30). Indications for PMRT in men are the same as those for PMRT in women (8). The NCCN and ASTRO guidelines are compared and summarized in the following section and in Tables 1 and 2. Unless otherwise specified, those recommendations based on nodal status apply to patients who have undergone full axillary LN dissection (ALND).

NCCN and ASTRO Guidelines

Four or More Positive Nodes at ALND.—Both the NCCN and the ASTRO provide guidelines for cases of four or more positive LNs found at ALND. The NCCN recommends that the chest wall, infraclavicular-supraclavicular area, IMNs, and any part of the axillary bed that is at risk be included for radiation (Fig 5). The ASTRO recommends including the same areas, with the exception of the IMNs and supraclavicular LNs, stating that there is insufficient evidence to recommend for or against radiation in patients who are clinically node negative (29) (Fig 6).

Tumors with One to Three Positive Nodes.—The ASTRO (29) states that there is insufficient evidence to routinely offer PMRT to patients who have T1–T2 tumors with one to three positive nodes, and the decision should be tailored to individual patient risks. Moreover, several studies (29,31,32) have shown local-regional recurrence of less than 10% of cases in this setting. Nevertheless, for T1–T3 tumors with one to three positive nodes, the NCCN describes this is as a “strongly consider” indication (Fig 7). This is an area of controversy in which clinicians can greatly benefit from discussions in MTBs. Factors to consider in the risk-benefit analysis include patient characteristics, tumor biologic factors, tumor burden, and presence of micrometastases (lesions >0.2 to ≤2 mm seen at sentinel node biopsy) (8,33). If radiation is indicated, the NCCN recommends including the chest wall, infra- and supraclavicular nodes, IMNs, and axillary bed (7). As noted previously, the ASTRO states that there is insufficient evidence to recommend for or against radiating IMNs and supraclavicular nodes (29).

Positive Sentinel Node Biopsy Results without ALND.—According to an informal consensus, the ASTRO recommends PMRT for patients with positive sentinel node biopsy results only if there is already sufficient information to justify the use of radiation without knowing the status of the intact axillary nodes. NCCN guidelines do not address this indication. However, for cases in which ALND is not performed and an LN micrometastasis is found, they do recommend that radiation be performed if the tumor is located in the central or medial region of the breast, or is larger than 2 cm and the patient is young or has extensive lymphovascular invasion. Accurate assessment of tumor size and localization within the breast are essential tasks that radiologists perform.

Axillary Nodes That Persist after NAST.—According to the ASTRO, PMRT should be considered when axillary nodes persist after the patient has undergone NAST. Per NCCN guidelines, in patients with operable disease that shows only a partial

Table 1: Indications for PMRT Based on NCCN and ASTRO Guidelines

Guideline	Indications Based on NCCN Guidelines*	Indications Based on ASTRO Guidelines
Recommend PMRT if		
	Four or more positive nodes One to three positive nodes Inflammatory breast cancer Recurrence Initially inoperable tumor T3 or stage III tumors with positive nodes Micrometastases and no ALND if: Tumor has central or medial location OR tumor > 2 cm, AND either young patient or extensive LVI	Four or more positive nodes T1–T3 disease with one to three positive nodes[†] Positive margins T3 or stage III tumors with positive nodes Extranodal extension
Strongly consider PMRT if		
	T1–T3 disease with one to three positive nodes Positive margins after re-excision Nodal involvement after NAST	Positive SNB results without ALND[‡]
Consider PMRT if		
	Large (>5 cm) tumor with negative axillary nodes Tumor ≤ 5 cm, margins negative but < 1 mm, negative nodes Tumor > 2 cm, or either central with young patient or extensive LVI Tumor < 5 cm; negative nodes; margins > 1 mm if tumor is central or >2 cm with at least 10 axillary LNs removed and at least one is grade 3, is estrogen receptor negative, or with LVI	Axillary nodal involvement after NAST

Note.—Comparisons were performed according to guidelines. ALND = axillary LN dissection, LVI = lymphovascular invasion, SNB = sentinel node biopsy.
* Applicable to invasive cancers specifically.
[†] PMRT recommended for this indication if the benefits outweigh the risks, with the patient's characteristics, tumor burden, and tumor biologic factors taken into account.
[‡] Only if there is already sufficient information to justify its use without needing to know that additional axillary nodes are involved (29).

response or progression after NAST, radiation to the chest wall, infra- and supraclavicular nodes, IMNs, and axillary bed should be strongly considered.

T3 or Stage III Cancer with Positive Axillary Nodes.—Both the ASTRO and the NCCN recommend PMRT in cases of T3 or stage III disease with positive axillary nodes.

Mastectomy for Inflammatory Breast Cancer and Response to Preoperative Systemic Therapy.—The ASTRO does not address cases in which mastectomy is performed for inflammatory breast cancer and the tumor responds to preoperative systemic therapy. However, the NCCN recommends radiation therapy to the chest wall, infraclavicular region, supraclavicular area, IMNs, and any part of the axillary bed that is at risk in such cases.

Negative Axillary Nodes and Tumor Larger than 5 cm.—The ASTRO does not address cases of negative axillary nodes with tumors larger than 5 cm. However, the NCCN recommends that PMRT be considered in these cases (Fig 8). The NCCN withholds a stronger recommendation for PMRT because there is evidence of generally low local-regional failure rates in patients in this disease group. In five National Surgical Adjuvant Breast and Bowel Project chemotherapy trials (34), local-regional recurrence as low as 7% was found among patients with pT3N0 disease who underwent mastectomy alone or mastectomy with adjuvant systemic therapy. If PMRT is planned, the fields to include are the chest wall, infra- and supraclavicular nodes, IMNs, and axillary bed.

Positive Surgical Margins.—In the setting of positive surgical margins, the ASTRO recommends PMRT and the NCCN advises that PMRT be strongly considered. Per the NCCN, the areas that should be irradiated include the chest wall, regional nodes, and undissected axilla.

Surgical Margins Narrower than 1 mm with 5-cm or Smaller Tumor and Negative Axillary Nodes.—The ASTRO does not provide guidelines for cases in which the surgical margins are less than 1 mm in width, the tumor is 5 cm or smaller, and the axillary nodes are negative for malignancy. However, the NCCN advises that PMRT be considered for this disease group.

Negative Margins, Negative Nodes, and Small Tumors.—The ASTRO does not provide guidelines for cases in which the tumor margins and nodes are negative and the tumor is smaller than 2 cm. However, the NCCN advises that PMRT be considered if the tumor is central in location or larger than 2 cm with at least 10 axillary nodes removed and at least one is grade 3, is estrogen receptor negative, or has lymphovascular invasion.

Local-Regional Recurrence.—For local-regional recurrence, the ASTRO provides no guidelines, as compared with the

Table 2: NCCN and ASTRO PMRT Guidelines Based on Disease Group

Indication	NCCN Guideline	ASTRO Guideline
Inflammatory breast cancer	Recommend PMRT	...
LNs (T1–3, M0)*		
Four or more positive axillary nodes	Recommend PMRT	Recommend PMRT
One to three positive axillary nodes	Strongly consider PMRT	Recommend PMRT
Micrometastases and no ALND	Recommend PMRT if: Tumor has central or medial location *OR* tumor > 2 cm *AND* either patient is young or there is extensive LVI	...
No positive axillary nodes, tumor ≤ 5 cm, margins negative or < 1 mm	PMRT usually not recommended Consider PMRT for: Central or medial tumors < 2 cm, fewer than 10 LNs excised; *AND* tumor is grade 3 or estrogen receptor negative or with LVI	...
No positive axillary nodes and tumor > 5 cm	Consider PMRT	...
Any positive axillary nodes and tumor > 5 cm	Recommend PMRT	Recommend PMRT
Extranodal extension	NA	Recommend PMRT
Margins positive at re-excision	Strongly consider PMRT	Recommend PMRT
DCIS: invasive cancer also found *AND* positive axillary nodes at ALND	Recommend PMRT	...
Response to NAST†		
Clinically positive nodes become clinically negative with NAST	Strongly consider PMRT	...
Partial response or progression with NAST *AND* positive nodes at ALND	Recommend PMRT	...
Partial response or progression with NAST *AND* clinically positive nodes	Strongly consider PMRT	...
Tumor initially inoperable but becomes operable after NAST	Recommend PMRT	...
Local recurrence		
Local recurrence and no prior radiation therapy	Recommend PMRT with surgery	...
Axillary node, supraclavicular node, and/or IMN recurrence	Recommend PMRT, if possible, with surgery	...
Pregnancy in first trimester, continued pregnancy elected	Recommend PMRT postpartum	...

Note.—Comparisons were performed according to patient group. DCIS = ductal carcinoma in situ, LVI = lymphovascular invasion, NA = not applicable.
* Based on findings at full ALND.
† Based on maximal stage at diagnosis (before NAST) and pathologic stage at definitive surgery (after NAST).

NCCN, which recommends administering PMRT following surgical excision.

Extensive Ductal Carcinoma in Situ Treated with Mastectomy.— Neither the ASTRO nor NCCN guidelines address cases of extensive ductal carcinoma in situ that is treated with mastectomy.

Further Considerations

The NCCN guidelines address the optimal order of trimodality therapy. Patients are usually treated with chemotherapy before radiation therapy, but there is no evidence of a detriment in using opposite sequencing, and it may be useful to initiate PMRT first if chemotherapy would otherwise be delayed for any reason. Certain chemotherapeutic agents such as cyclophosphamide, methotrexate, and fluorouracil can be given concurrently with radiation therapy, as can endocrine agents and adjuvant human epidermal growth factor receptor 2 –targeted therapy (35). However, in practice, chemotherapy and PMRT are typically administered sequentially.

Specific dosing practices and the areas to be included in the radiation therapy field are variable among guidelines and therefore among institutions. For example, the NCCN recommends a radiation boost to the mastectomy scar and drains in cases of inflammatory breast cancer but not in all PMRT cases. As another example, the ASTRO recommends treatment to the chest wall in patients who undergo PMRT, but it does not specify the total dose, use of bolus material, or targeting. These remain institution dependent and represent areas of further discussion outside the scope of this article.

Breast Reconstruction

Breast reconstruction following mastectomy is optional, is considered safe in patients who are eligible for surgery, and does not negatively impact the prognosis (8,36). At least half

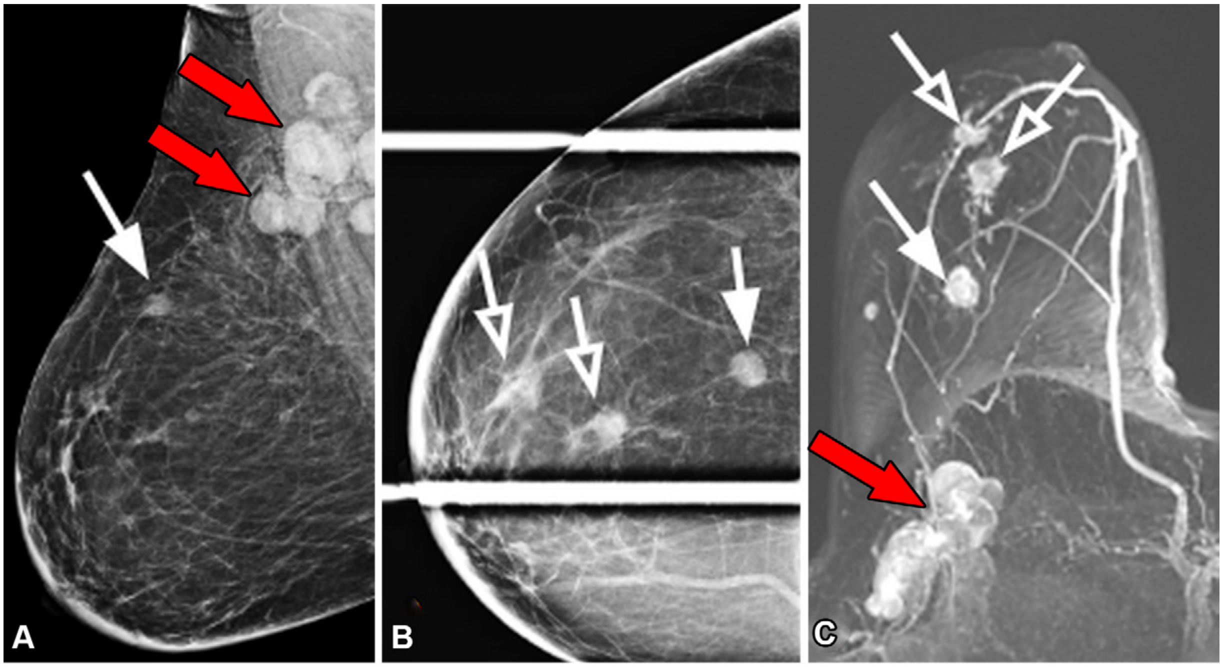


Figure 5. Palpable right breast lump in a 64-year-old woman. **(A, B)** Mediolateral oblique **(A)** and spot compression craniocaudal **(B)** mammograms show a mass (solid white arrow), two focal asymmetries (open arrows in **B**), and several morphologically abnormal axillary LNs (red arrows in **A**). **(C)** Axial maximum intensity projection MR image findings confirm the presence of three enhancing breast masses (solid and open white arrows), including the focal asymmetries (open arrows) seen in **B**, and axillary adenopathy (red arrow). **(D)** PET/CT scan (attenuated correction view) shows at least three fluorodeoxyglucose-avid axillary LNs (arrows). The patient underwent neoadjuvant chemotherapy, followed by modified radical mastectomy with ALND and immediate saline implant reconstruction. Pathologic analysis revealed nuclear grade 3 estrogen receptor–, progesterone receptor–, human epidermal growth factor receptor 2–negative invasive ductal carcinoma with micropapillary features. Fourteen of 15 axillary LNs, several with extranodal extension, were positive for metastatic disease. This is an example of a “recommended” indication for PMRT, as more than three axillary LNs were positive at ALND. The patient underwent 50-Gy radiation in 25 fractions to the chest wall and regional LNs and completed the treatment as planned. She developed grade 2 dermatitis in the right chest wall.

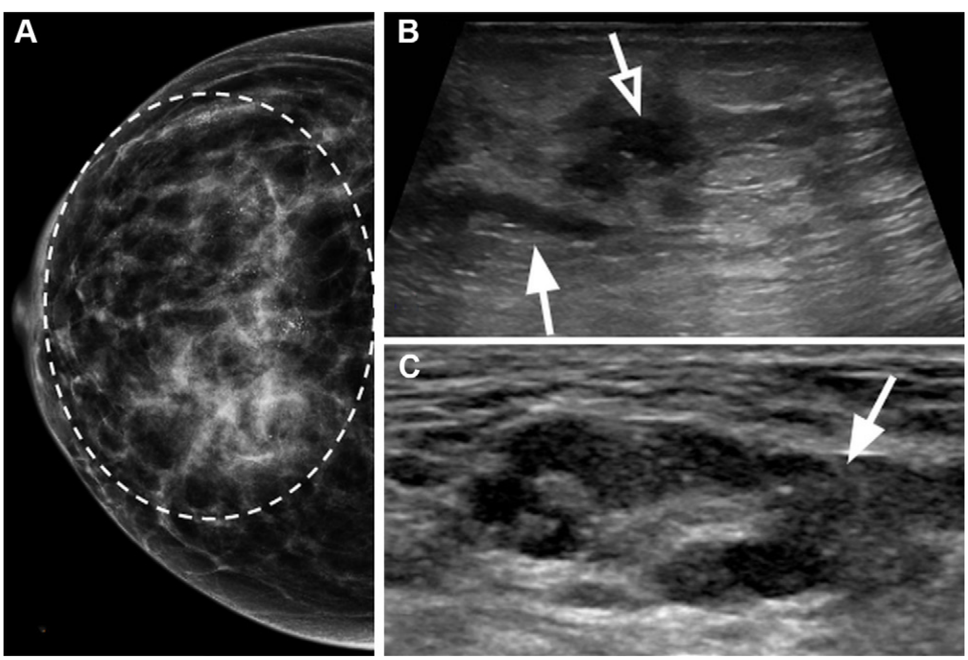


Figure 6. No symptoms in a 56-year-old woman. **(A) Right craniocaudal** mammogram shows extensive pleomorphic calcifications (oval outline) in a regional distribution. **(B)** Corresponding US image shows dilated ducts (open arrow) with intraluminal calcifications and solid material (solid arrow). **(C)** US image of the ipsilateral axilla shows a morphologically abnormal LN with areas of hypoechoic cortical thickening (arrow). The patient underwent modified radical mastectomy and ALND and deferred undergoing reconstruction. Pathologic analysis revealed invasive and in situ ductal carcinoma. The tumor measured 4.5 cm, and eight of 16 axillary LNs were metastatic. This is an example of a “recommended” indication for PMRT per both NCCN and ASTRO guidelines, as more than three positive axillary LNs were found at ALND. The patient underwent 50-Gy radiation in 25 fractions to the right chest wall and regional nodes. She completed the treatment despite developing hyperpigmentation and desquamation in the right chest wall.

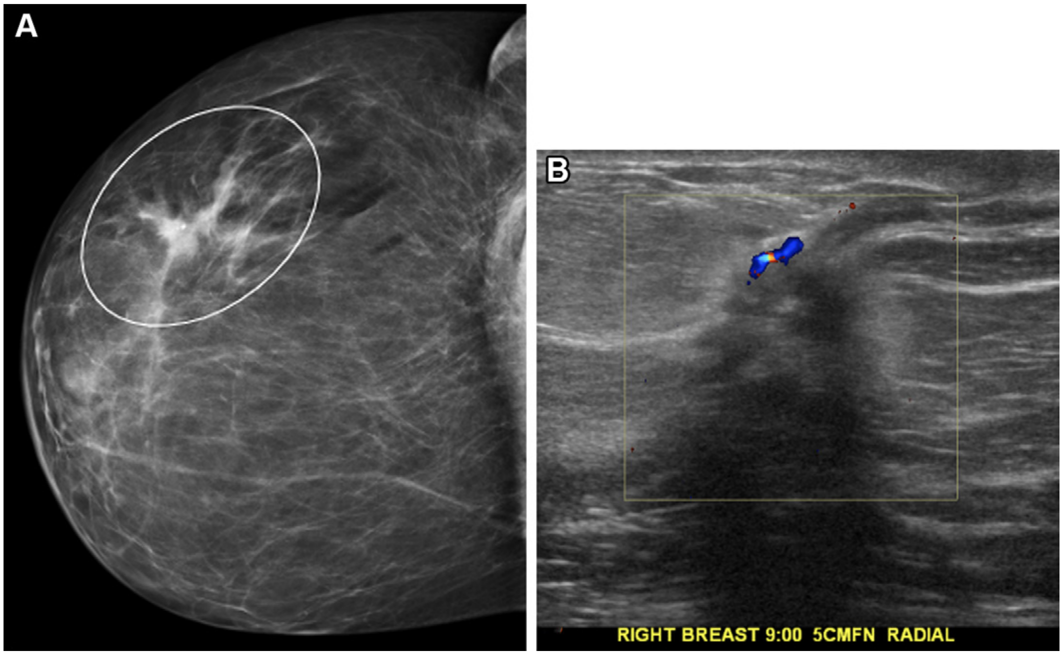


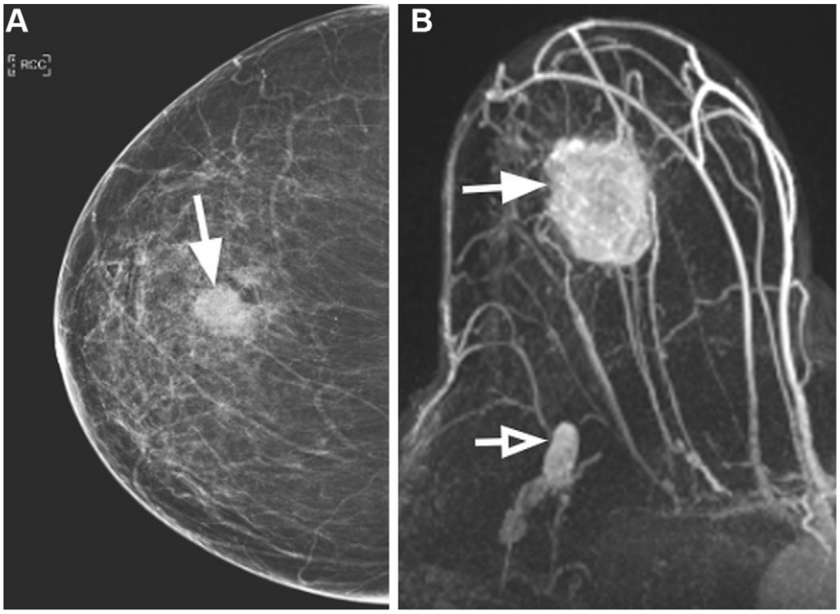
Figure 7. No symptoms in a 42-year-old woman with a history of ovarian cancer. **(A)** Mediolateral mammogram shows a developing asymmetry (oval outline) in the upper right breast. **(B)** Color Doppler US image of an area 5 cm from the nipple shows a nonparallel, irregular, hypoechoic mass with spiculated margins and posterior shadowing. Subsequent biopsy yielded grade 2 invasive mammary carcinoma. The patient underwent mastectomy and ALND. The tumor measured 2.4 cm, and two of five LNs were metastatic with extranodal extension. The patient elected to undergo bilateral mastectomy and NAST. This is an example of a "strongly consider" indication for PMRT, as the patient is young with a positive axillary LN at ALND. She underwent 50-Gy radiation in 25 fractions to the chest wall and regional nodes. Delayed two-stage implant reconstruction was eventually performed.

of patients who undergo mastectomy decide to undergo breast reconstruction, which has been shown to improve patients' emotional and sexual well-being (2). However, those patients who undergo PMRT in addition to reconstruction are at increased risk of postoperative complications associated with reconstruction (8). Therefore, when PMRT is contemplated, the choice of the type and timing of breast reconstruction becomes complicated. The method ultimately used depends on multiple factors. One of the most important considerations is patient preference, which cannot always be accommodated because there are competing concerns such as surgical expertise, institutional preference, extent of disease, anticipation of adjuvant or neoadjuvant chemotherapy, and patient anatomy and comorbidities. Cross-disciplinary coordination during an MTB enables the care team to achieve a balance between the best reconstruction results and the fewest complications (37).

Breast reconstruction methods include autologous, implant-related, and combined autologous–implant-related approaches. With autologous reconstruction, the patient's own abdominal, buttock, or thigh tissue, including the skin, fat, and axial blood supply from the donor site, is used to recreate a breast mound. This transfer of tissue, with its native blood supply, is called a flap. This reconstruction may be immediate—that is, performed at the time of mastectomy—or delayed—that is, performed after completion of mastectomy and adjuvant chemotherapy. Recently, several academic centers have initiated a hybrid "delayed-immediate" approach in which a tissue expander is placed at the time of mastectomy, with PMRT and finally autologous flap surgery following (38).

Traditionally, autologous reconstruction has been favored over implant reconstruction in patients who receive PMRT because it results in fewer complications (2). For example, in a review by El-Sabawi et al (39), it was reported that radiation therapy administered in patients with autologous reconstructions resulted in a reconstruction failure rate of 7%, while Ho et al (37) described a reconstruction failure rate of 18.8% in patients with implant-related reconstructions after radiation therapy. Moreover, autologous reconstruction tends to provide better cosmetic results (39), particularly if the contralateral breast is extremely ptotic. However, there may be related complications, including volume loss, breast deformity, and fat necrosis. Smoking, poorly controlled diabetes mellitus, and obesity confer an increased risk of autologous reconstruction–related complications, likely owing to microvascular disease, and are considered relative contraindications (8). Moreover, autologous reconstruction is complex for both surgeons and patients: Few training programs offer instruction regarding this surgery, and many patients are not healthy enough to endure the procedure. This may explain why 80% of all postmastectomy reconstructions are performed by using implant-based reconstructions (2).

With implant-related reconstruction, a prosthetic breast composed of saline and/or silicone is implanted to create a breast mound. The implant may be placed immediately, at the time of mastectomy. Alternatively, a two-step (delayed) approach can be used. This approach entails inserting a temporary saline-filled tissue expander at the time of mastectomy, gradually inflating the expander with saline in the physician's



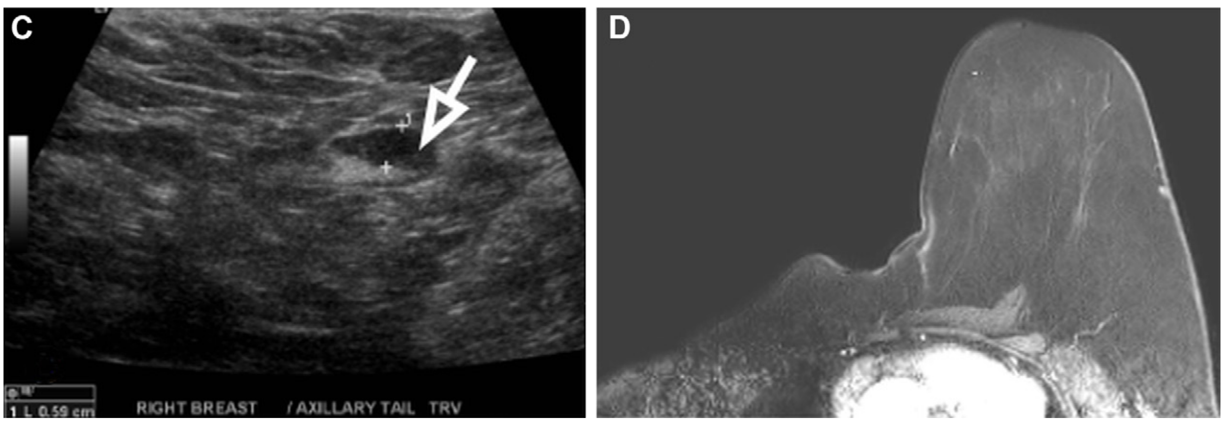


Figure 8. Palpable mass in the right breast in a 51-year-old woman. **(A)** Right craniocaudal mammogram shows an irregular high-density mass (arrow) in the central breast. **(B)** Axial maximum intensity projection MR image shows a 4.5-cm irregular enhancing mass (solid arrow) in the central breast and eccentric cortical thickening (open arrow) of an axillary LN. **(C)** US image of the right axilla shows nodal cortical thickening (arrow) measuring 6 mm. Pathologic analysis revealed a 6.5-cm invasive ductal carcinoma with medullary features. The LNs were negative for metastatic disease at ALND. The patient underwent NAST followed by modified radical mastectomy. This is an example of a "consider" indication for PMRT, as the tumor is large and located centrally but with negative axillary LNs at ALND. The patient underwent 46/50-Gy radiation in 23/25 fractions to the chest wall and regional LNs. PMRT was stopped owing to grade 3 dermatitis to allow skin healing. It was determined that the patient had received a sufficient radiation dose, and the final two radiation therapy sessions were omitted. **(D)** Contrast-enhanced breast surveillance MR image after NAST shows no abnormal or suspicious enhancement.

office over several months, and exchanging the expander with an implant in a later surgery. A hybrid approach normally involves combining the transferal of a latissimus muscle and tissue flap to the mastectomy space, with placement of a tissue expander or implant behind it. This type of reconstruction is usually reserved as a salvage option after failed initial reconstruction (2). Common complications of implant-related reconstruction in the setting of PMRT include infection, capsular contracture, and extrusion of the implant through the skin.

The optimal method and timing of implant-related reconstruction are controversial. At our institution, if it is known before surgery that the patient needs PMRT, our plastic surgeons believe that the best option is to delay the reconstruction or perform a flap reconstruction. However, often, PMRT is not planned until after the mastectomy because the final determined disease dictates whether it is indicated. If immediate reconstruction is chosen, many microsurgeons consider autologous reconstruction to be the primary choice for patients with clinical stage 2 or 3 disease (Fig 9) because patients have tolerated radiation better with this approach than previously suspected (40). The best alternative to autologous reconstruction is to place a tissue expander at the time of mastectomy, radiate the expander, and then wait 4–6 months after the radiation therapy before replacing the expander with an implant (40). This allows the surgeon to properly assess the skin after radiation therapy.

In short, the best approach to reconstruction in the setting of PMRT has been highly debated, largely because of the paucity of robust data (2,8). If PMRT is planned, the NCCN recommends that radiation therapy precede autologous reconstruction. If implant-related reconstruction is chosen, the NCCN recommends using the two-step tissue expander–implant approach (8).

Radiation Toxicity–related Complications

Complications from radiation therapy may be acute or chronic and may not be seen until decades after the treatment. Complications range from mild dermatitis to rare but potentially fatal radiation-induced second malignancies. They may be classified as early (occurring weeks to months after radiation therapy), intermediate (occurring months to years after radiation therapy), or late (occurring more than 10 years after radiation therapy). Radiation-associated toxic effects may be augmented by certain systemic agents. For example, doxorubicin increases the risk of toxic effects in the heart when administered concurrently with PMRT (29). The harmful effects of chemotherapeutic agents also may be additive. For example, doxorubicin increases the risk of cardiac disease when used in combination

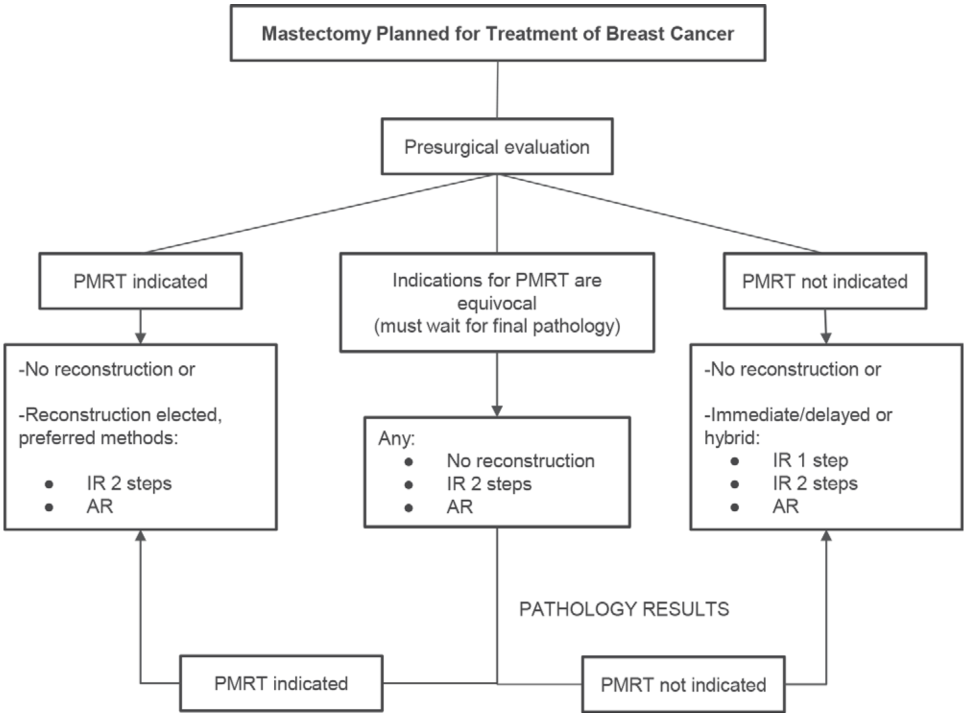


Figure 9. Diagram illustrates the steps involved in breast reconstruction for breast cancer management. *AR* = autologous reconstruction, *IR* = implant-related reconstruction.

with trastuzumab. Last, human epidermal growth factor receptor 2–directed therapy and anthracyclines administered alone may cause left heart dysfunction (41) and lead to congestive heart failure (42). Therefore, cardiac dysfunction in patients with breast cancer who have undergone chemotherapy and radiation therapy is often the result of compounded effects.

As noted, a potentially deleterious complication of breast radiation therapy is cardiac disease. Darby et al (43) estimated the risk of cardiac disease among women who were treated for breast cancer between 1958 and 2001. They estimated the average mean dose of radiation to the heart to be greater with left-sided tumors (6.6 Gy) than with right-sided tumors (2.9 Gy). The dose was estimated to be even higher (10 Gy) in patients with left-sided tumors and a small thoracic wall who received radiation therapy to the IMNs. The rate of major coronary events increased by 7.4% per 1 Gy of increased radiation dose ($P < .001$), with no apparent minimal threshold necessary to increase the cardiac risk. The increase in coronary events started within the first 5 years after radiation therapy and continued into the 3rd decade after the treatment. The proportional increase in the rate of major coronary events per gray was similar between women with and those without cardiac risk factors at the time of radiation therapy.

Radiation pneumonitis may manifest acutely with ground-glass opacities or consolidation. Subacute pneumonitis (seen in 1%–4% of patients) occurs more often in patients who receive radiation therapy to regional LNs and/or concurrently with chemotherapy. It is seen in up to 20% of patients who are treated with both radiation therapy and paclitaxel (44,45). Radiation may also result in late pulmonary fibrosis (Fig 10).

Radiation-related skin effects include acute edema, which occurs as an inflammatory response that results in increased vascular permeability and radiation-induced dermatitis. Acute radiation-induced dermatitis usually is seen within 90 days after radiation therapy. It can manifest as erythema as early as within 24 hours after treatment and usually resolves in a few days. Later, around weeks 2–4, persistent erythema with dryness and hyperpigmentation may be seen. After this, if the cumulative dose reaches 20 Gy, moist desquamation might appear. If the dose reaches 40 Gy, therapy might need to be interrupted until re-epithelialization occurs.

Chronic radiation-induced dermatitis occurs months or years after treatment. More chronic skin manifestations include telangiectasia, dermal atrophy, pigmentation, fibrosis, edema, keratosis, and even dermal necrosis (46). These conditions typically resolve after months but may persist for years after treatment and are seen as skin thickening. Chronic fluid collections also may develop (Fig 11) and be difficult to differentiate from recurrence or radiation-induced sarcomas of the chest wall. In these cases, biopsy may be needed. Fluid collections and deep soft-tissue masses may be clinically occult and are examples of findings that are likely to be first detected by radiologists. Another frequent complication of radiation therapy is lymphedema, which may be potentiated by other factors such as increased patient body mass index, the extent or number of nodes dissected, radiation dose, and the use of a posterior axillary radiation field.

Radiation-induced osteonecrosis (Fig 12), an uncommon complication, occurs owing to vascular compromise with obliterative endarteritis and damage to osteoblasts and osteoclasts (10). To induce osteonecrosis, a dose greater than 6 Gy in adults is required. The onset of radiation-induced osteonecrosis occurs more than 1 year after treatment (47). In addition, there is an increased risk of bone sarcomas in the irradiated field, with a latency period longer than 5 years.

Other late complications include brachial plexopathy and radiation-induced second malignancy (Fig 13). Radiation-induced second malignancies include angiosarcomas,

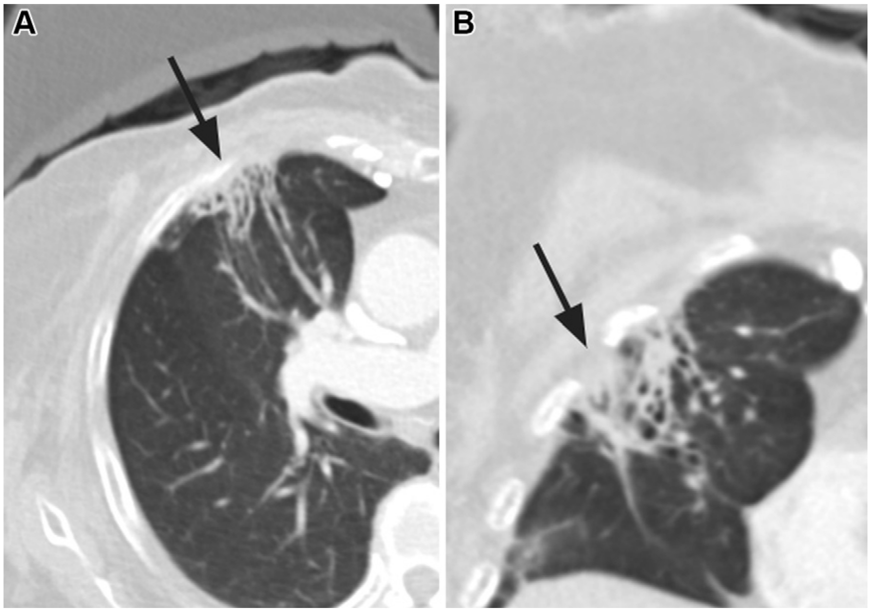


Figure 10. Findings after PMRT in a woman with a history of invasive mammary carcinoma. Axial **(A)** and sagittal **(B)** chest CT images show linear pulmonary opacities (arrow) in the irradiated field, with a somewhat well-demarcated triangular shape, which is typical for pulmonary fibrosis caused by radiation. There is also associated mild traction bronchiectasis.

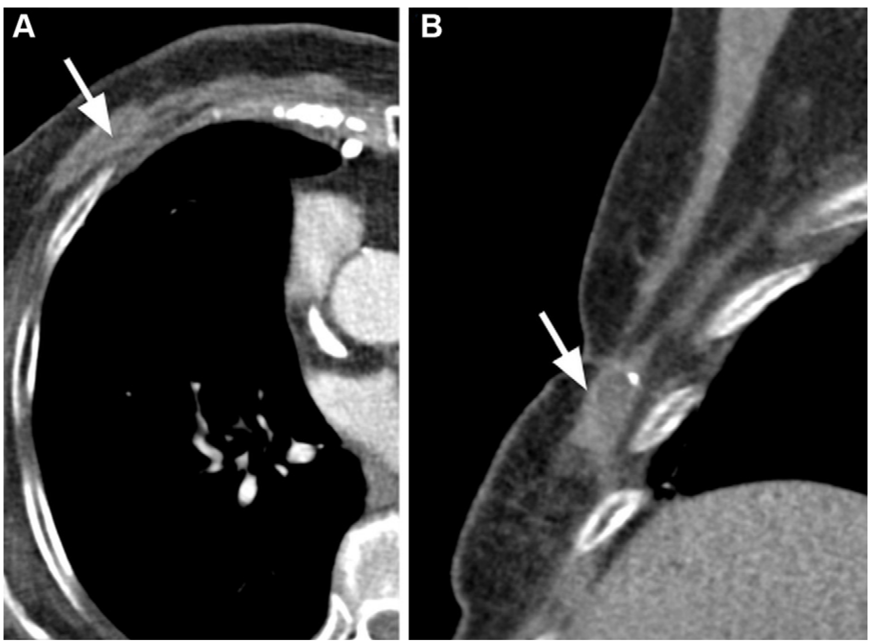


Figure 11. Findings after PMRT in a woman with a history of invasive mammary carcinoma. Axial **(A)** and sagittal **(B)** chest CT images show a fluid collection (arrow) in the anterior chest wall, immediately anterior to a rib, associated with a coarse calcification. This might reflect posttreatment changes; however, in the setting of prior malignancy, recurrence is almost always an alternative diagnosis. Owing to suspicion for recurrence, the patient underwent US-guided core biopsy, which revealed only a chronic hematoma.

pleomorphic sarcomas, invasive ductal carcinomas, and lymphomas. A brief summary of toxicities and possible differential diagnoses is provided in Table 3 (48–50). Scrutinizing imaging studies for signs of secondary malignancy and recurrent disease is an important radiologist contribution in the surveillance of patients with breast cancer.

Conclusion

Radiation therapy is generally administered after lumpectomy, but in some cases, it is administered even following mastectomy. PMRT can play an important role in breast cancer treatment by decreasing local-regional recurrence and improving overall survival. The use of PMRT can impact the type and timing of breast reconstruction. NCCN and ASTRO guidelines for the use of PMRT are continually evolving, and the decision to use this treatment is best made at MTBs, which breast imagers attend (Fig 14). Radiologists who are familiar with the indications for and complications of mastectomy and postmastectomy radiation can contribute essential information to important clinical discussions and help guide decisions regarding the most appropriate and safest patient management.

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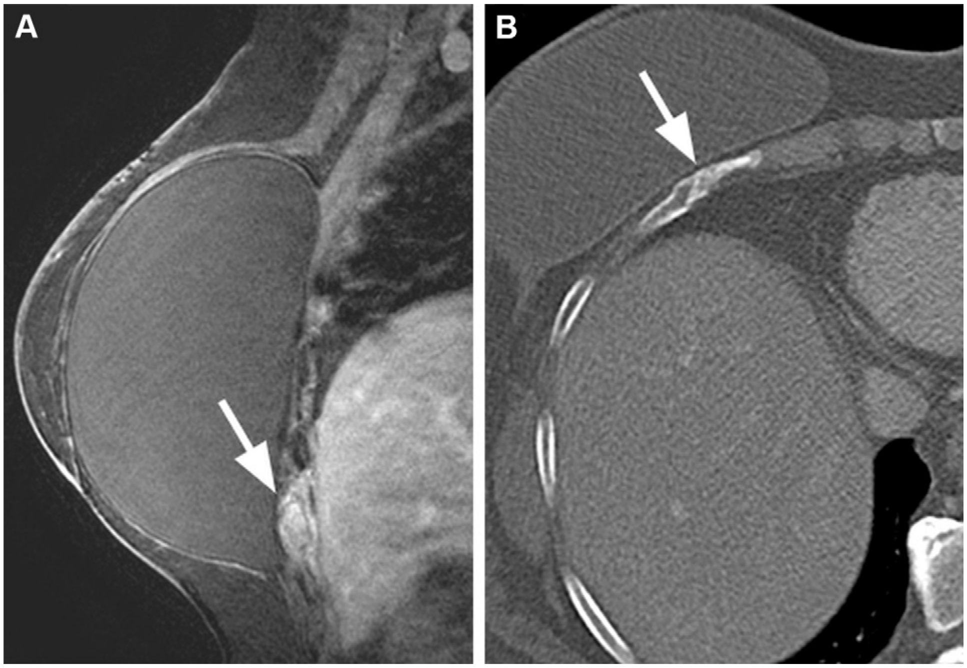

Figure 12. Findings in a woman with a history of PMRT and implant reconstruction who presented with pain in the inframammary fold. Owing to concern for recurrence, breast MRI was performed. **(A)** Sagittal contrast-enhanced T1-weighted MR image with fat saturation shows enhancement of a rib (arrow) subjacent to the implant, and, thus, chest CT was recommended. **(B)** Axial chest CT image shows an acute, minimally displaced fracture (arrow) of the right 7th rib. The fraction was believed to be the result of osteonecrosis from radiation therapy, as the affected rib was in the radiation therapy field and no other fractures were identified.

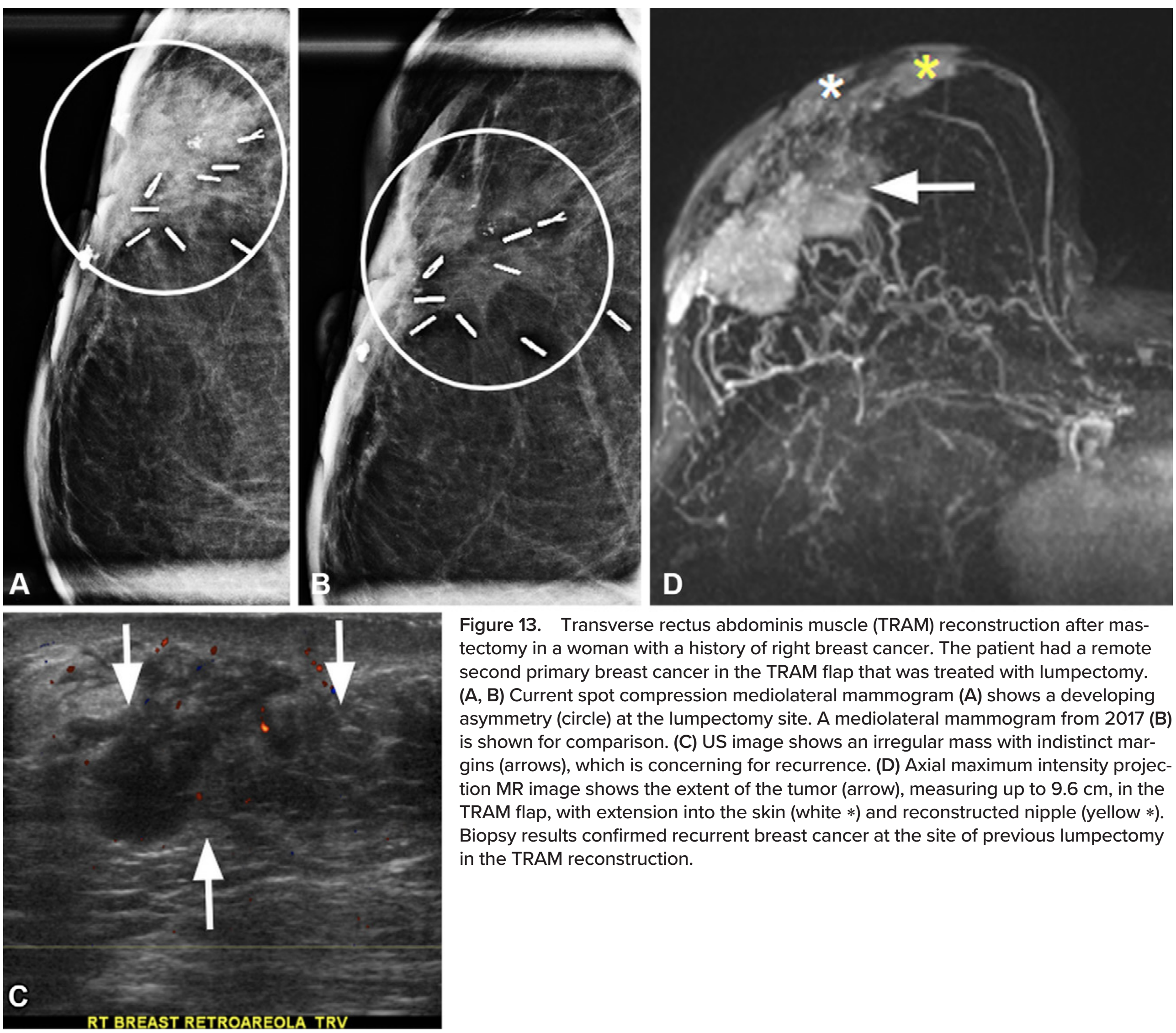

Figure 13. Transverse rectus abdominis muscle (TRAM) reconstruction after mastectomy in a woman with a history of right breast cancer. The patient had a remote second primary breast cancer in the TRAM flap that was treated with lumpectomy. **(A, B)** Current spot compression mediolateral mammogram **(A)** shows a developing asymmetry (circle) at the lumpectomy site. A mediolateral mammogram from 2017 **(B)** is shown for comparison. **(C)** US image shows an irregular mass with indistinct margins (arrows), which is concerning for recurrence. **(D)** Axial maximum intensity projection MR image shows the extent of the tumor (arrow), measuring up to 9.6 cm, in the TRAM flap, with extension into the skin (white ∗) and reconstructed nipple (yellow ∗). Biopsy results confirmed recurrent breast cancer at the site of previous lumpectomy in the TRAM reconstruction.

Table 3: Manifestations of Radiation Toxicity after Radiation Therapy for Breast Cancer

Most Common Location	Subjective Physical Examination Findings	Radiologic Findings	Differential Diagnosis
Skin	Acute: erythema, pain, dryness, ulceration, swelling Chronic: desquamation, thickening	Inflammatory changes in the subcutaneous fat Skin thickening, trabeculation	Cellulitis Dermatitis of other causes (ie, atopic) Chronic: angiosarcoma
Mammary gland	Volume loss	Glandular atrophy	Age-related atrophy
Lungs and pleura	Acute and chronic: dyspnea	Acute: ground-glass opacity, consolidation within or outside (rare) radiated field, seen in at least 62% of cases (48) Chronic: reticular opacities with or without bronchiectasis, volume loss within radiated field Pleural effusions may develop 6 months after therapy	Pneumonia, organizing pneumonia Pulmonary edema for other reasons Fibrosis due to other reasons (ie, connective tissue disease) Malignant pleural effusion, which might require cytologic analysis
Radiation-induced malignancy	Skin thickening (angiosarcoma)	Soft tissue within irradiated field (pleomorphic sarcoma)	Dermatitis, recurrence
Osteonecrosis	Chest pain, point tenderness in chest wall	Fracture within radiated field with or without bone scan showing sharply defined local increased uptake	Age-related osteoporosis Traumatic fracture
Radiation-induced cardiac disease	Dyspnea, chest pain Systolic dysfunction is more common than diastolic dysfunction (49)	Pericardial effusion, constrictive pericarditis Coronary artery disease	Other causes of pericardial effusion Atherosclerotic disease
Brachial plexus injury	Occurs in 1.2% of patients with breast cancer after radiation therapy (50)	Abnormal soft-tissue thickening or nodularity along the brachial plexus	Tumor infiltration or recurrence

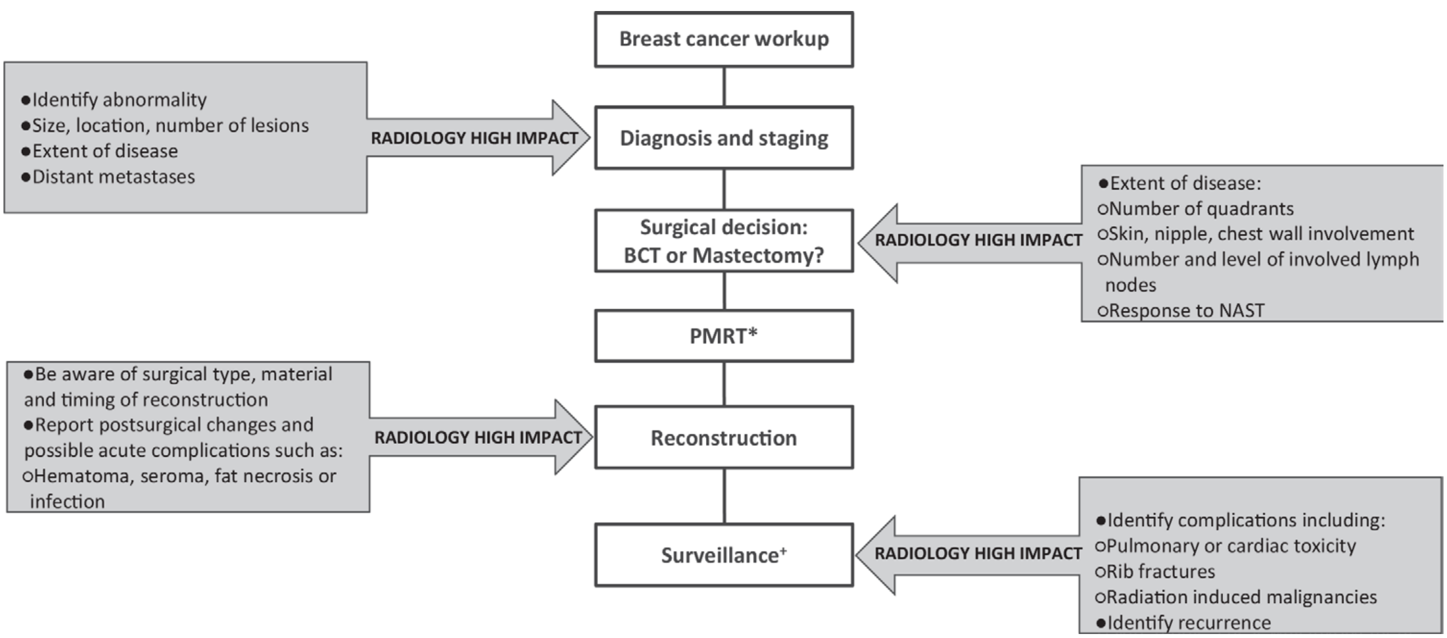

Figure 14. Diagram outlines radiologists' high-impact contributions to MTBs, with a focus on clinical issues related to PMRT (*). †During surveillance, an oncology consultation should be recommended if suspicious imaging findings are identified.

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